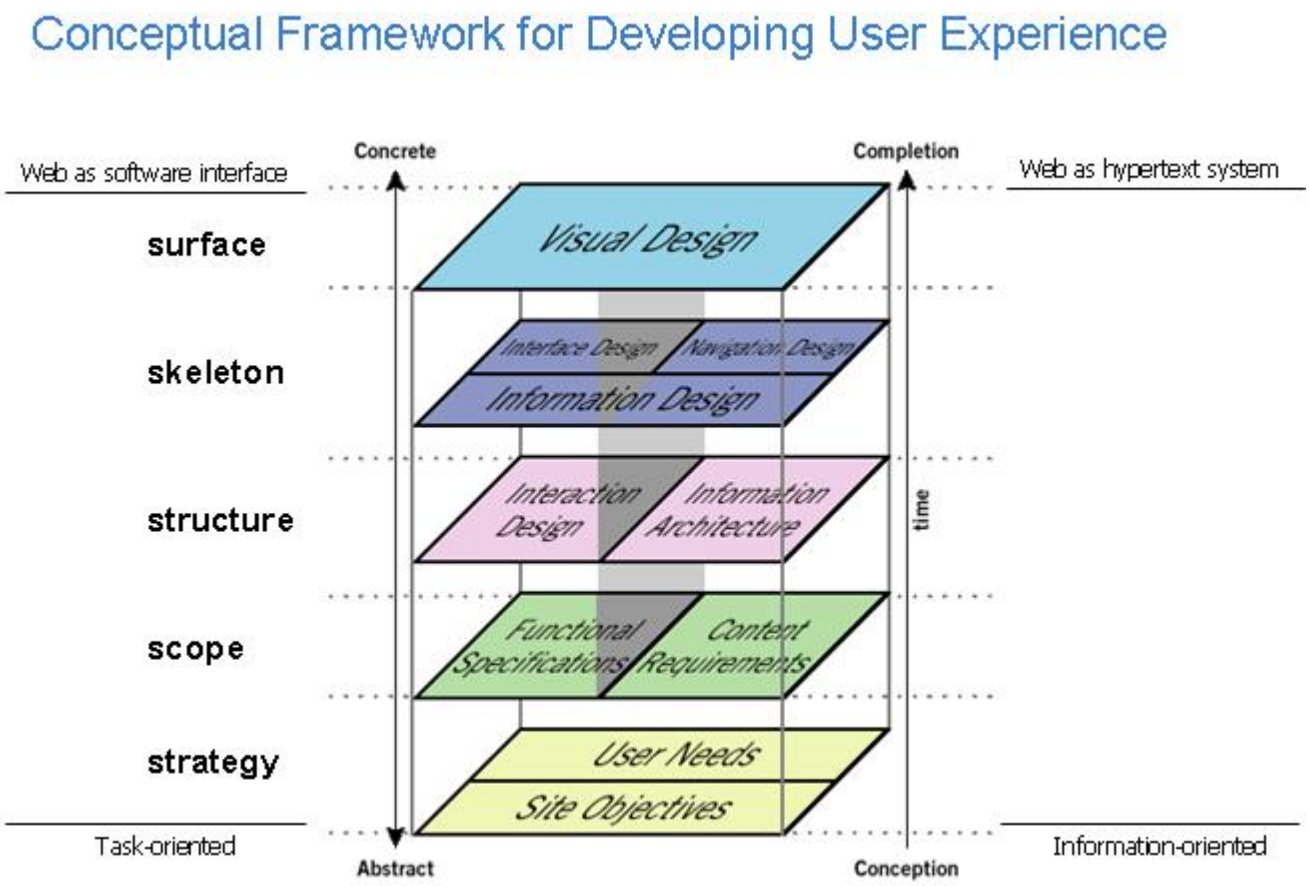
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Conceptual framwork name



Definition of HCI

"Human-Computer Interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of **major phenomena surrounding them"**

Goals of HCI

The term Human Computer Interaction (HCI) was adopted in the mid-1980s as a means of describing this new field of study. This term acknowledged that the focus of interest was broader than just the design of the interface and was concerned with all those aspects that relate to the interaction between users and computers.

The goals of HCI are to produce usable and safe systems, as well as functional systems. These goals can be summarized as 'to develop or improve the safety, utility,

effectiveness, efficiency and usability of systems that include computers' (Interacting with computers, 1989).

Reasoning

Reasoning is the process by which we use the knowledge we have to draw conclusions or infer something new about the domain of interest. There are a number of different types of reasoning:

- Deductive reasoning
- Inductive reasoning
- Abdicative reasoning

Deductive reasoning

Deductive reasoning derives the logically necessary conclusion from the given premises. For example,

It is Friday then she will go to work
It is Friday
Therefore she will go to work

It is important to note that this is the logical conclusion from the premises; it does not necessarily have to correspond to our notion of truth. So, for example,

If it is raining then the ground is dry

Inductive reasoning

Induction is generalizing from cases we have seen to infer information about cases we have not seen. For example, if every elephant we have ever seen has a trunk, we infer that all elephants have trunks. Of course, this inference is unreliable and cannot be proved to be true; it can only be proved to be false. We can disprove the inference simply by producing an elephant without a trunk.

Abdicative reasoning

The third type of reasoning is abduction. Abduction reasons from a fact to the action or state that caused it. This is the method we use to derive explanations for the events we observe. For example, suppose we know that Sam always drives too fast when she has been drinking. If we see Sam driving too fast we may infer that she has been drinking. Of course, this too is unreliable since there may be another reason why she is driving fast: she may have been called to an emergency, for example.

In spite of its unreliability, it is clear that people do infer explanations in this way and hold onto them until they have evidence to support an alternative theory or explanation. This can lead to problems in using interactive systems.

Possible user persona type designations include:

- Primary: the persona's needs are sufficiently unique to require a distinct interface form and behavior
- Secondary: primary interface serves the needs of the persona with a minor modification or addition
- Supplement: the persona's needs are fully satisfied by a primary interface
- Served: the persona is not an actual user of the product, but is indirectly affected by it and its use
- Negative: the persona is created as an explicit, rhetorical example of whom not to design for

Evaluations done during design to check that the product continues to meet users' needs are known as **formative evaluations**. Evaluations that are done to assess the success of a finished product, such as those to satisfy a sponsoring agency or to check that a standard is being upheld, are known as **summative evaluation**. Agencies such as National Institute of Standards and Technology (NIST) in the USA, the International Standards Organization (ISO) and the British Standards Institute (BSI) set standards by which products produced by others are evaluated.

Personas and user-centered design

Personas also resolve three User-Centered design issues that arise during product development:

• The elastic user
• Self-referential design
• Design edge cases

The elastic user

Although satisfying the user is goal, the term user causes trouble when applied to specific design problems and contexts. Its imprecision makes it unusable as a design tool—every person on a product team has his own conceptions of the user and what the user needs. When it comes time to make a product decisions, this "user" becomes elastic, bending and stretching to fit the opinions and presuppositions of whoever has the floor.

Self-referential design

Self-referential design occurs when designers or developers project their own goals, motivations, skills, and mental models onto a product's design. Most "cool" product designs fall into this category: the audience doesn't extend beyond people like the designer, which is fine for a narrow range of products and completely inappropriate for most others. Similarly, programmers apply self-referential design when they create implementation-model products.

Design edge cases

Another syndrome that personas help prevent is designing for edge cases—those Situations that might possibly happen, but usually won't for the target personas. Naturally, edge cases must be programmed for, but they should never be the design focus. Personas Provide a reality check for the design.

Personas

To create a product that must satisfy a broad audience of users, logic tells you to make it as broad in its functionality as possible to accommodate the most people. This logic, however, is flawed. The best way to successfully accommodate a variety of users is to design for specific types of individuals with specific needs.

When you broadly and arbitrarily extend a product's functionality to include many constituencies, you increase the cognitive load and navigational overhead for all users. Facilities that map please some users will likely interfere with the satisfaction of other.

A simple example of how personas are useful is shown in figure below, if you try to design an automobile that pleases every possible driver, you end up with a car with every possible feature, but which pleases nobody. Software today is too often designed to please to many users, resulting in low user satisfaction

Navigation of information

Navigation of information, or of the content of panes or windows, can be accomplished

by several methods: scrolling (panning), linking (jumping), and zooming. The first two methods are common: scrolling is ubiquitous in most software and linking is ubiquitous on the Web (though increasingly, linking idioms are being adopted in non-Web applications). Zooming is primarily used for visualization of 3D and detailed 2D data. Scrolling is often a necessity, but the need for it should be minimized when possible. Often there is a tradeoff between paging and scrolling information: You should understand your users' mental models and workflows to determine what is best for them. In 2D visualization and drawing applications, vertical and horizontal scrolling is common. These kinds of interfaces benefit from a thumbnail map to ease navigation. Linking is the critical navigational paradigm of the Web. Because it is a visually dislocating activity, extra care must be taken to provide visual and textual cues that help orient users.

Q.No.1 it has been observed that most computer users use menu option for input instead of keyboard accelerator. What is the reason behind it? (Page#127)

Answer:-

1. Menu options are easier to find.

2. You don't have to memories the keys for menu option but for key board accelerators you have to memories them

Q.No.2 Define active intervention. (Page#276)

Answer:-

Active intervention with the participant and actively probes the participant understands of whatever is being tested.

Q.No.3 What is Ubiquitous Computing? (Page#140)

Answer:-

The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it. These

words have inspired a new generation of researchers in the area of ubiquitous computing.

Q.No.4 Which display device is very cheap and has fast enough response time for rapid animation with high color compatibility? (Page#119)

Answer:-

The CRT is a cheap display device and has fast enough response times for rapid animation coupled with a high color capability.

Q.No.5 By breakdown Skelton plane into its component elements then how all piece fit together to create this as whole user experience? (Page#306)

Answer:-

Beneath that surface is the skeleton of the site: the placement of buttons, tabs, photos, and blocks of text. The skeleton is designed to optimize the arrangement of these elements for maximum effect and efficiency

Q.No.6 Explain Meta data in term of data processing. (Page#406)

Answer:-

In data processing, meta-data is definitional data that provides information about or documentation of other data managed within an application or environment

Q.No.7 Suppose you are a system analyst, with your work experience, briefly describe when it is useful to observe in contrast to observation process? (Page#361)

Answer:-

Being a system analyst, Early in design, observation helps designers understand users' needs. Other types of observation are done later to examine whether the developing prototype meets users' needs.

Q.No.8 Describe any five usability goals of Internet Explorer. (Page#31)
Answer:-

Usability is broken down into the following goals:

1. Effective to use (effectiveness)
2. Efficient to use (efficiency)
3. Safe to use(safety)
4. Have good utility (utility)
5. Easy to learn (learnability)

1. Effectiveness:
It is a very general goal and refers to how good a system at doing what it is suppose to do.

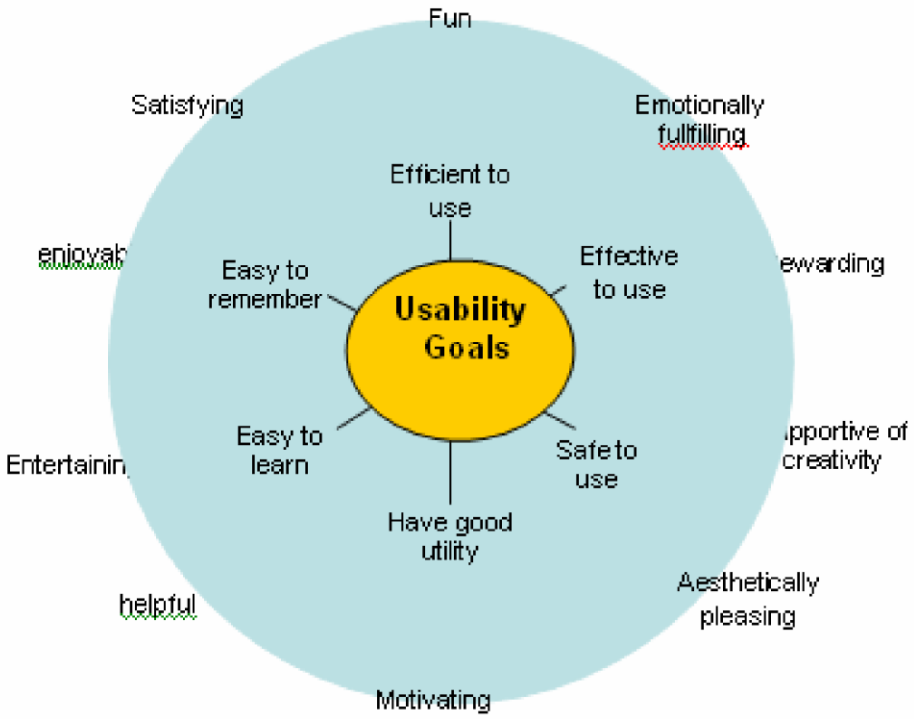
2. Efficiency:
It refers to the way a system supports users in carrying out their tasks.

3. Safety:
It involves protecting the users from dangerous conditions and undesirable situations. In relation to the first ergonomics aspect, it refers to the external conditions where people work.

4. Utility:
It refers to the extent to which the system provides the right kind of functionality so that user can do what they need or want to do.

5. Learnability:

It refers to how easy a system is to learn to use. It is well known that people do not like spending a long time learning how to use a system



Q.No.9 When observation is useful and how to observe? (Page#361)

Answer:-

Observing is useful at any time during product development. Early in design, observation helps designers understand users' needs.

How to observe:

The same basic data-collection tools are used for laboratory and field studies (i.e., direct observation, taking notes, collecting video, etc.) but the way in which they are used is different. In the laboratory the emphasis is on the details of what individuals do, while in the field the context is important and the focus is on how people interact with each other, the technology, and their environment. Furthermore, the equipment in the laboratory is usually set up in advance and is relatively static

whereas in the field it usually must be moved around. In this section we discuss how to observe, and then examine the practicalities and compare data-collection tools.

Q.No.10 Difference between Semi-structure interview and focus group interview (Page#373)

Answer:-

Semi-Structure Interview	Focus Group Interview
Semi-structured, interviews combine features of structured and unstructured inter views and use both closed and open questions.	One form of group interview is the focus group that is frequently used in marketing, political campaigning, and social sciences research.
Use of pre-determined questions provides uniformity	Normally three to 10 people are involved.
Requires some level of training or practice in order to prevent interviewer suggesting answers	Focus groups are also attractive because they are low-cost, provide quick results, and can easily be scaled to gather more data
It also allows respondents to discuss and raise issues that you may not have considered	A preset agenda is developed to guide the discussion

Q.No.11 Which theory claimed Problem solving is both productive and reproductive, what are controlled vocabularies? (Page#91)

Answer:-

Gestalt theory:

Gestalt psychologists were answering the claim, made by behaviorists, that problem solving is a matter of reproducing known responses or trial and error. This explanation was considered by the Gestalt school to be insufficient to account for human problem solving behavior.

Q.No.12 Point out the problem with Excise task? (Page#246)

Answer:-

The problem with excise tasks is that the effort we expend in doing them doesn't go directly towards accomplishing our goals. Where we can eliminate the need for excise tasks, we make the user more effective and productive and improve the usability of the software.

Q.No.13 Differentiate between Visual Art and Visual Design?

Answer:-

Visual Art	Visual Design
Visual art is art such as paintings, photographs, sculptures. The term visual art separates those kinds of art from music, dance, and drama. It is two-dimensional. Visual arts are things like paintings, drawings, visual designs, photography, and computer art. There is no "designing" for a purpose; it's just creating something visually appealing.	Visual design is the way a piece of artwork, a room, yard, video game, photo or other artistic subject is composed and appeals to the human eye. Photoshop designs for each page, page type, module within an app/site/software. Creates or contributes to style guides detailing how everything looks and how it fits into the larger strategy.

Q.No.14 How active intervention technique performed? (Page#276)

Answer:-

Active intervention is particularly useful early in design. It is an excellent technique to use with prototypes, because it provides a wealth of diagnostic information. It is not the technique to use; however, if your primary concern is to measure time to complete tasks or to find out how often users will call the help desk. To do a useful active intervention test, you have to define your goals and concerns, plan the questions you will use as probes, and be careful not to bias participants by asking leading questions.

Q.No.15 Application software implemented in file system model (Page#330)

Answer:-

Software implemented in the file system model: Our software is already designed and built that way, and users are used to it neither of these arguments is valid. The first one is irrelevant because new programs written with a unified file model can freely coexist with the older implementation model applications. The underlying file system doesn't change at all. In much the same way that toolbars quickly invaded the interfaces of most applications in the last few years, the unified file model could also be implemented with similar success and user acclaim.

The second argument is more insidious, because its proponents place the user community in front of them like a shield. What's more, if you ask users themselves, they will reject the new solution because they abhor change, particularly when that change affects something they have already worked hard to master — like the file system. However, users are not always the best predictors of design successes, especially when the designs are different from anything they've already experienced,

Q.No.16 Explain metadata in term of data processing (Page#406)

Answer:-

In data processing, meta-data is definitional data that provides information about or documentation of other data managed within an application or environment.
For example:-

Meta-data would document data about data elements or attributes (name, size, data type, etc) and data about records or data structures (length, fields, columns, etc) and data about data (where it is located, how it is associated, ownership, etc.).

Q.No.17 Define controlled Vocabularies? (Page#407)

Answer:-

Vocabulary control comes in many shapes and sizes. At its most vague, a controlled vocabulary is any defined subset of natural language. At its simplest, a controlled vocabulary is a list of equivalent terms in the form of a synonym ring, or a list of preferred terms in the form of an authority file.

Q.No.18 Write principles to prevent user privacy? (Page#268)

Answer:-

Most previous evaluations in human-computer interaction, these studies can be done without users knowing that they are being studied. This raises ethical concerns, chief among which are issues of privacy, confidentiality, informed consent, and appropriation of others' personal stories (Sharf, 1999). People often say things online that they would not say face to face.

Q.No.19 What is Scope Plane in context of elements of user experience? (Page#308)

Answer:-

On the software side, the strategy is translated into scope through the creation of functional specifications: a detailed description of the "feature set" of the product. On the information space side, scope takes the form of content requirements: a description of the various content elements that will be required. Now we can map that whole confusing array of terms into the model. By breaking each plane down into its component elements, we'll be able to take a closer look at how all the pieces fit together to create the whole user experience.

Q.No.20 By breaking down Skeleton plane into its component elements then how all the pieces fit together to create this as a whole user experience? (Page#308)

Answer:-

The skeleton plane breaks down into three components.

1. On both sides, we must address information design:
The presentation of information in a way that facilitates understanding.

2. For software products:
The skeleton also includes interface design, or arranging interface elements to enable users to interact with the functionality of the system.

3. The interface for an information space is its navigation design:
The set of screen elements that allow the user to move through the information architecture.

Q.No.21 What "function" will create a copy of the original document in MS Word? (Page#326)

Answer:-

The word snapshot makes it clear that the copy is identical to the original, while also making it clear that the copy is not tied to the original in any way. That is, subsequent changes to the original will have no effect on the copy.

Q.No.22 Define Strategy and Scope planes? (Page#307,308)

Answer:-

Strategy Planes:
The scope is fundamentally determined by the strategy of the site. This strategy incorporates not only what the people running the site want to get out of it but what the

users want to get out of the site as well. In the case of our bookstore example, some of the strategic objectives are pretty obvious: Users want to buy books, and we want to sell them. Other objectives might not be so easy to articulate.

Scope planes:
On the software side, the strategy is translated into scope through the creation of functional specifications: a detailed description of the "feature set" of the product. On the information space side, scope takes the form of content requirements: a description of the various content elements that will be required.

Q.No.23 Usability tests? (Page#275)

Answer:-

In most usability tests, you have one participant at a time working with the product. You usually leave that person alone and observe from a corner of the room or from behind a one-way mirror.

Q.No.24 Identify the main components that make up a global navigation system in context of a web. (Page#287,288)

Answer:-

Web designers use the term penitent navigation (or global navigation) to describe the set of navigation elements that appear on every page of a site, Persistent navigation should include the five elements you most need to have on hand at all times.

1. The Home page.
2. Forms.
3. Site ID
4. The Sections
5. The Utilities

Q.No.25 You are a software engineer and design different applications that are usable for users without modifications. How do you define accessibility with respect to emerging paradigm? (Page#408)

Answer:-

Accessibility is a general term used to describe the degree to which a system is usable by as many people as possible without modification. It is not to be confused with usability which is used to describe how easily a thing can be used by any type of user. One meaning of accessibility specifically focuses on people with disabilities and their use of assistive devices such as screen-reading web browsers or wheelchairs.

Q.No.26 Suppose you are an HCI specialist, what do you think that evaluation of design or product is beneficial? Justify your point of view. (Page#21,257)

Answer:-

HCI specialists test design ideas on real users and use formal evaluation techniques to replace intuition in guiding design. This constant reality check improves the final product.
Most effectively develop programs and products designed to meet customer needs, all operations are strongly science based. For example, our research and development teams emphasize improving rearing methodologies for current and new products, designing consistently significant quality control evaluations, improving natural enemy delivery methodologies and developing technologies for field population monitoring. Our science-based efforts are the best means to achieve and maintain the highest standards of program and product quality.

1. Problems are fixed before the product is shipped, not after.
2. The team can concentrate on real problems, not imaginary ones.
3. Engineers code instead of debating.
4. Time to market is sharply reduced.

5. Finally, upon first release, your sales department has a rock-solid design it can sell without having to pepper their pitches with how it will all actually work in release 1.1 or 2.0.

Q.No.27 Match the following regarding Goal-Directed Design

A	B
Research	Users and use Context
Modeling	Definition of user, business& technical needs
Requirements	Definition of design structure & flow
Framework	Of behavior, form & content
Refinement	User and the Domain

Answer:-

A	**B**
Research	User and the Domain
Modeling	Users and use Context
Requirements	Definition of user, business& technical needs
Framework	Definition of design structure & flow
Refinement	Of behavior, form & content

Q.No.28 Suppose you are creating an account on yahoo, fill all the required information but forget to retype password. As you click on submit button a warning is message appears "This information is required" on retype password field. Yahoo suggests that your registration will not be submitted until you retype password. Explain this specific characteristic of considerate software.

Answer:-

It ensures there are no mistakes while typing the password because retyping matches and makes sure you do not make a mistake. You can type an error password once but not twice. This is a security check and it's good for user.

Q.No.29 You are assigned to perform trunk test on website "amazon.com". Write down all the necessary steps to perform the trunk test. (Page#296)

Answer:-

Imagine that you've been blindfolded and locked in the trunk of a car, then driven around for a while and dumped on a page somewhere deep in the bowels of a Web site. If the page is well designed, when your vision clears you should be able to answer these questions without hesitation:

1. What site is this? (Site ID) **amazon.com**
2. What page am I on? (Page name)
3. What are the major sections of this site? (Sections)
4. What are my options at this level? (Local navigation)
5. Where am I in the scheme of things? ("You are here" indicators) **6.** How can I search?

Q.No.30 I am working on a web site and have designed a home page. The information I added does not fit into screen size of computer. I do not want to remove any data from this page. Is this any solution that all data can be shown to user?

Answer:-

Tabs can be used to group related data under one heading and then creating different sections. Scroll bars can be used for data which does not fit on one screen. Different sections can be created with a heading and some content given on main page with a "Read More" indicator. This will take user to a new page if user is interested in reading more.

Q.No.31 Suppose you are a web developer. While designing web applications you use metadata tags in your application that make efficient search? Briefly explain metadata and its purpose in web site. (Page#406)

Answer:-

A web site is a collection of interconnected systems with complex dependencies. A single link on a page can simultaneously be part of the site's structure, organization, labeling, navigation, and searching systems. It's useful to study these systems independently, but it's also crucial to consider how they interact. Reductionism will not tell us the whole truth.

Q.No.32 Suppose you are an HCI specialist and have to perform the usability testing for those products that do not even have to involve any hardware or software. Which usability testing techniques you will use in this case. (Page#259)

Answer:-

Usability testing was the dominant approach in the 1980s (Whiteside et al., 1998), and remains important, although, as you will see, field studies and heuristic evaluations have grown in prominence.

Q.No.33 Which screen display when user launches the program and first loads into the memory? (Page#398)

Answer:-

A splash screen is a dialog box displayed when a program first loads into memory. Sometimes it may just be the about box or Identity box, displayed automatically, but more often publishers create a separate splash screen that is more engaging and visually exciting

Q.No.34 What is the mean by life cycle of module? Why they are used in software development. (Page#149)

Answer:-

Understanding what activities are involved in interaction design is the first step to being able to do it, but it is also important to consider how the activities are related to one another so that the full development process can be seen. The term lifecycle

model is used to represent a model that captures a set of activities and how they are related.
SDLC stands for Software Development Life Cycle. A Software Development Life Cycle is essentially a series of steps, or phases, that provide a model for the development and lifecycle management of an application or piece of software. The methodology within the SDLC process can vary across industries and organizations, but standards such as ISO/IEC 12207 represent processes that establish a lifecycle for software, and provide a mode for the development, acquisition, and configuration of software systems

Q.No.35 Explain the concept of Attentive environment? (Page#418)

Answer:-

Attentive environments are environments that are user and context aware. One project which explores these themes is IBM's Blue Eyes research project is chartered to explore and define attentive environments, software and provide a mode for the development, acquisition, and configuration of software systems

Q.No.36 Explain the following planes of conceptual framework for development interactive product? (Page#306,307)

- The structure plane
- The strategy plane

Answer:-

The structure plane: The skeleton is a concrete expression of the more abstract structure of the site. The skeleton might define the placement of the interface elements on our checkout page; the structure would define how users got to that page and where they could go when they were finished there. The skeleton might define the arrangement of navigational items allowing the users to browse categories of books; the structure would define what those categories actually were.

The strategy plane: The scope is fundamentally determined by the strategy of the site. This strategy incorporates not only what the people running the site want to get out of it but what the users want to get out of the site as well. In the case of our bookstore example, some of the strategic objectives are pretty obvious: Users want to buy books, and we want to sell them. Other objectives might not be so easy to articulate.

Q.No.37 What are good reasons for investing in user testing which are points out by tognazzini? (Page#257)

Answer:-

Tognazzini points out that there are five good reasons for investing in user testing:

1) Problems are fixed before the product is shipped, not after.
2) The team can concentrate on real problems, not imaginary ones.
3) Engineers code instead of debating.
4) Time to market is sharply reduced.
5) Finally, upon first release, your sales department has a rock-solid design it can sell without having to pepper their pitches with how it will all actually work in release 1.1 or 2.0.

Q.No.38 How navigation excised justify your answer? (Page#248)

Answer:-

The most important thing to realize about navigation is that, in almost all cases, it represents pure excise, or something close to it. Except in games where the goal is to navigate successfully through a maze of obstacles, navigating through software does not meet user goals, needs, or desires. Unnecessary or difficult

Q.No.39 Why some time direct observation is not possible? (Page#366)

Answer:-

Sometimes direct observation is not possible because it is obtrusive or evaluators cannot be present over the duration of the study, and so users' activities are tracked indirectly.

Q.No.40 Give two examples of "Avoidance" with respect to Attitude-oriented questions. (Page#183)

Answer:-

What would you prefer not to do? What
do you procrastinate on?

Q.No.41 Write any two common ways to Improve Navigation? (Page#408)

Answer:-

1. One integrated within a web site or intranet to improve navigation and retrieval, shares a common heritage with the familiar reference text but has a different form and function.

2. Thesaurus is a semantic network of concepts, connecting words to their synonyms, homonyms, antonyms, broader and narrower terms, and related terms.

Q.No.42 List down any four principles that can help make your visual interface as easy and pleasurable to use as possible. (Page#345)

Answer:-

1) Avoid visual noise and clutter
2) Use contrast, similarity, and layering to distinguish and organize elements
3) Provide visual structure and flow at each level of organization
4) Use cohesive, consistent, and contextually appropriate imagery **5)** Integrate style and function comprehensively and purposefully

Q.No.43 Explain "Participants Represent Real Users" in context of usability testing. (Page#270)

Answer:-

While there can be wide variations in where and how you conduct a usability test, every usability test shares these five characteristics:

1. The primary goal is to improve the usability of a product. For each test, you also have more specific goals and concerns that you articulate when planning the test.

2. The participants represent real users.

3. The participants do real tasks.

4. You observe and record what participants do and say.

5. You analyze the data, diagnose the real problems, and recommend changes to fix those problems.

Q.No.44 When is it useful to observe in context to observation process? (Page#361)

Answer:-

Observing is useful at any time during product development. Early in design, observation helps designers understand users' needs. Other types of observation are done later to examine whether the developing prototype meets users' needs

Q.No.45 Discuss the model problems of multiple undo. (Page#314)

Answer:-

The problems with multiple undo are not due to its behavior as much as they are due to its manifest model. Most undo facilities are constructed in an unrelentingly

function-centric manner. They remember what the user does function-by-function and separate the user's actions by individual function. In the time-honored way of creating manifest models that follow implementation models, undo systems tend to model code and data structures instead of user goals. Each click of the Undo button reverses precisely one function-sized bite of behavior. Reversing on a function-by-function basis is a very appropriate mental model for solving most simple problems caused by the user making an erroneous entry. Users sense it right away and fix it right away, usually within a two- or three-function Limit.

For Example: The Paint program in Windows 95 had a fixed, three-action undo limit. However, when the problem grows more convoluted, the incremental, multiple undo models don't scale up very well.

Q.No.46 How can we improve data retrieval through "storage and retrieval Systems"? (Page#402)

Answer:-

A storage system is a method for safekeeping goods in a repository. It is a physical system composed of a container and the tools necessary to put objects in and take them back out again. A retrieval system is a method for finding goods in a repository. It is a logical system that allows the; goods to be located according to some abstract value, like name, position or some aspect of the; contents

Q.No.47 Does it is necessary for evaluation team to have the expertise needed to do the evaluation? Justify your answer with example. (Page#402)

Answer:-

Does the evaluation team have the expertise needed to do the evaluation? **Example,** if no one has used models to evaluate systems before, then basing an evaluation on this approach is not sensible. It is no use planning to use experts to review an interface if none are available. Similarly, running usability tests requires expertise. Analyzing video can take many hours, so someone with appropriate expertise and equipment must be available to do it. If statistics are to be used, then a statistician should be consulted before starting the evaluation and then again later for analysis, if appropriate.

Q.No.48 Define reliability in term of evaluation technique. (Page#268)

Answer:-

The reliability or consistency of a technique is how well it produces the same results on separate occasions under the same circumstances. Different evaluation processes have different degrees of reliability.

Q.No.49 What is the first step which describes the process of defining the interaction framework? (Page#205)

Answer:-

DEFINING FORM FACTOR AND INPUT METHODS

QNo.50 What is meant by term "Color shows relationships" with respect to misuse of colors in visual Interface? (Page#358)

Answer:-

Color shows relationships. Color can provide a means of grouping or relating objects together.

QNo.51 Explain any four disadvantages of Low-fidelity prototyping.

Answer:-

1. Limited error checking.
2. Poor detailed specification to code to.
3. Facilitator-driven.
4. Limited utility after requirements established.

QNo.52 Explain any two necessary conditions regarding "Page Names"? (Page#290)

Answer:-

1. Street signs are big. When you're stopped at an intersection, you can read the sign for the next cross street.
2. They're in the right place—hanging ovsr the street you're driving on, so all you have to do is glance up.

Q.No.53 Can we make computers "see" and "feel"?

Answer:-

We'd have to understand how the human body processes these things. Seeing is the easier of the two, but unless you are talking of some tactile sensor, I don't know how someone would make a computer feel.

Q.No.54 Describe quantitative data analysis. (Page#368)

Answer:-

Video data collected in usability laboratories is usually annotated as it is observed Small teams of evaluator's watch monitors showing what is being recorded in a control room out of the users' sight. As they see errors or unusual behavior, one of the evaluators marks the video and records a brief remark. When the test is finished evaluators can use the annotated recording to calculate performance times so they can compared users' performance on different prototypes

Q.No.55 Explain the following in context of problems with the web experience:
No sense of scale.
No sense of location.
(Page#283,284)

Answer:-

No sense of scale:
Even after we've used a Web site extensively, unless it's a very small site we tend to have very little sense of how big it is (50 pages? 1,000? 17,000?). For all we know, there could be huge corners we've never explored. Compare this to a magazine, a museum, or a department store, where you always have at least a rough sense of the seen/unseen ratio. The practical result is that it's very hard to know whether you've seen everything of interest in a site, which means it's hard to know when to stop looking.

No sense of location:
In physical spaces, as we move around we accumulate knowledge about the space. We develop a sense of where things are and can take shortcuts to get to them. We may get to the chainsaws the first time by following the signs, but the next time we're just as likely to think,

Q.No.56 Explain the following planes of Conceptual Framework for developing an interactive product. The Structure Plane The Scope Plane The Surface Plane The Strategy Plane
(Page#306,307)
Answer:-

The Structure Plane:
The skeleton is a concrete expression of the more abstract structure of the site. The skeleton might define the placement of the interface elements on our checkout page; the structure would define how users got to that page and where they could go when they were finished there. The skeleton might define the arrangement of navigational items allowing the users to browse categories of books; the structure would define what those categories actually were.

The Scope Plane:
The structure defines the way in which the various features and functions of the site fit together. Just what those features and functions are constitutes the scope of the site. Some sites that sell books offer a feature that enables users to save previously used addresses so they can be used again. The question of whether that feature-or any feature-is included on a site is a question of scope.

The Surface Plane:

On the surface you see a series of Web pages, made up of images and text. Some of these images are things you can click on, performing some sort of function such as taking you to a shopping cart. Some of these images are just illustrations, such as a photograph of a book cover or the logo of the site itself.

The Strategy Plane:
The scope is fundamentally determined by the strategy of the site. This strategy incorporates not only what the people running the site want to get out of it but what the users want to get out of the site as well. In the case of our bookstore example, some of the strategic objectives are pretty obvious: Users want to buy books, and we want to sell them. Other objectives might not be so easy to articulate

Q.No.57 Explain the term "Accessibility" with respect to Emerging Paradigms.

Answer:-

Accessibility is a general term used to describe the degree to which a system is usable by as many people as possible without modification. It is not to be confused with usability which is used to describe how easily a thing can be used by any type of user. One meaning of accessibility specifically focuses on people with disabilities and their use of assistive devices such as screen-reading web browsers or wheelchairs.

Q.No.58 Define excise? (Page#245)

Answer:-

When we decide to drive to the office, we must open the garage door, get in, start the motor, back out, and close the garage door before we even begin the forward motion that will take us to our destination. All these actions are in support of the automobile rather than in support of getting to the destination.

Q.No.59 Different between evaluation and usability? (Page#259,264)

Answer:-

Evaluation	Usability
Focus on users and their tasks Observe, measure, and analyze their performance with the system Design lucratively	**Efficiency:** How much time, and how many steps, are required for people to complete basic tasks **Accuracy:** How many mistakes did people make **Recall:** How much does the person remember afterwards or after periods of non-us **Emotional response:** How does the person feel about the tasks completed

Q.No.60 Name types of navigation? (Page#249)

Answer:-

- Navigation between multiple windows or screens
- Navigation between panes within a window (or frames in a page)
- Navigation between tools or menus in a pane
- Navigation within information displayed in a pane or frame (for example: scrolling, panning, zooming, following links)

Q.No.61 What are Ancillary Application Windows? (Page#396)

Answer:-

Ancillary application windows are windows that are not really part of the application's functionality, but are provided as a matter of convention

Q.No.62 How data needs are different from functional needs? (Page#204)

Answer:-

Functional needs are the operations that need to be performed on the objects of the system and which are eventually translated into interface controls. Functional needs also define places or containers where objects or information in the interface must be displayed.

Q.No.63 What is the widespread misconception about the error messages? (Page#382)

Answer:-

This is a widespread misconception. Most error message boxes are informing the user of the inability of the program to work flexibly. Most error message boxes seem to the user like an admission of real stupidity on the program's part.

Q.No.64 List three characteristics that must be present in a good website.

Answer:-

Good Keywords: Good keywords are the very first building blocks you must consider before your webpage even becomes a dim notion in your head

Simple Design: Keep it simple. You must keep your webpage simple and direct. Keep it professional

Easy Navigation: A great webpage will have easy and simple navigation.

Q.No.65 How visual explicitness of the GUI helps casual or first-time user to navigate and learn what tasks are appropriate and when? How can we relate excise with expert users? (Page#246)

Answer:-

One of the main criticisms leveled at graphical user interfaces by experienced computer users — notably those trained on command-line systems — is that getting to where you want to go is made slower and more difficult by the extra effort that

goes into manipulating windows and icons. Users complain that, with a command line, they can just type in the desired command and the computer executes it immediately. With windowing systems, they must open various folders looking for the desired file or program before they can launch it. Then, after it appears on the screen, they must stretch and drag the window until it is in the desired location and configuration.

Any user willing to learn a command-line interface automatically qualifies as a power user. And any power user of a command-line interface will quickly become a power user of any other type of interface, GUI included. These users will easily learn each nuance of the programs they use. They will start up each program with a clear idea of exactly what it is they want to do and how they want to do it. To this user, the assistance offered to the casual or first-time user is just in the way.